ExamGOAL
PYQ PDF

Free · With answer key

NEET · BIOLOGY · CHAPTER-WISE PYQS

Mineral Nutrition

NEET Biology · Previous Year Questions with Answer Key

62	39	1989 – 2023	62	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Out of syllabus	62 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Out of syllabus 62 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q1. NEET 2023 Manipur · MCQ (Single Correct Answer)

Which of the following mineral ion is not remobilized in plants?

(A) Potassium (B) Calcium (C) Nitrogen (D) Phosphorus

Open solution

Q2. NEET 2023 Manipur · MCQ (Single Correct Answer)

Match Column I with Column II.

	Column - I		Column - II
(A)	Nitrococcus	(I)	Denitrification
(B)	Rhizobium	(II)	Conversion of ammonia to nitrite
(C)	Thiobacillus	(III)	Conversion of nitrite to nitrate
(D)	Nitrobacter	(IV)	Conversion of atmospheric nitrogen to ammonia

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(B) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(C) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
(D) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Open solution

Q3. NEET 2023 · MCQ (Single Correct Answer)

Which micronutrient is required for splitting of water molecule during photosynthesis?

(A) Molybdenum (B) Magnesium (C) Copper (D) Manganese

Open solution

Q4. NEET 2023 · MCQ (Single Correct Answer)

Match List I with List II:

	List I		List II
(A)	Iron	(I)	Synthesis of auxin
(B)	Zinc	(II)	Component of nitrate reductase
(C)	Boron	(III)	Activator of catalase
(D)	Molybdenum	(IV)	Cell elongation and differentiation

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I
(B) A-III, B-I, C-IV, D-II
(C) A-II, B-IV, C-I, D-III
(D) A-III, B-II, C-I, D-IV

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. NEET 2022 Phase 2 · MCQ (Single Correct Answer)

Which of the following protects nitrogenase inside the root nodule of a leguminous plant?

(A) Glutamate dehydrogenase
(B) Catalase
(C) leg haemoglobin
(D) Transaminase

Open solution

Q6. NEET 2022 Phase 1 · MCQ (Single Correct Answer)

Match List-I with List-II

	List - I		List - II
(a)	Manganese	(i)	Activates the enzyme catalase
(b)	Magnesium	(ii)	Required for pollen germination
(c)	Boron	(iii)	Activates enzymes of respiration
(d)	Iron	(iv)	Functions in splitting of water during photosynthesis

Choose the correct answer from the options given below:

(A) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(B) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(C) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(D) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Open solution

Q7. NEET 2022 Phase 1 · MCQ (Single Correct Answer)

Which one of the following produces nitrogen fixing nodules on the roots of Alnus?

(A) Rhizobium
(B) Frankia
(C) Rhodospirillum
(D) Beijerinckia

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. NEET 2021 · MCQ (Single Correct Answer)

Match Column - I with Column - II.

Open solution

	Column-I		Column-II
(a)	Nitrococcus	(i)	Denitrification
(b)	Rhizobium	(ii)	Conversion of ammonia to nitrite
(c)	Thiobacillus	(iii)	Conversion of nitrite to nitrate
(d)	Nitrobacter	(iv)	Conversion of atmospheric nitrogen to ammonia

Choose the correct answer from options given below.

(A)

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

(B)

(a)	(b)	(c)	(d)
(ii)	(iv)	(i)	(iii)

(C)

(a)	(b)	(c)	(d)
(i)	(ii)	(iii)	(iv)

(D)

(a)	(b)	(c)	(d)
(iii)	(i)	(iv)	(ii)

Q9. NEET 2020 Phase 2 · MCQ (Single Correct Answer)

In *Glycine max*, the product of biological nitrogen fixation is transported from the root nodules to other parts as :

(A) Ureides (B) Ammonia (C) Glutamate (D) Nitrates

Open solution

Q10. NEET 2020 Phase 2 · MCQ (Single Correct Answer)

Which of the following elements helps in maintaining the structure of ribosomes ?

(A) Molybdenum (B) Magnesium (C) Zinc (D) Copper

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q11. NEET 2020 Phase 1 · MCQ (Single Correct Answer)

Match the following concerning essential elements and their functions in plants

(a) Iron (i) Photolysis of water
(b) Zinc (ii) Pollen germination
(c) Boron (iii) Required for chlorophyll biosynthesis
(d) Manganese (iv) IAA biosynthesis

Select the correct option

(A) (A) → (iv), (B) → (iii), (C) → (ii), (D) → (i)
(B) (A) → (iv), (B) → (i), (C) → (ii), (D) → (iii)
(C) (A) → (iii), (B) → (iv), (C) → (ii), (D) → (i)
(D) (A) → (ii), (B) → (i), (C) → (iv), (D) → (iii)

Open solution

Q12. NEET 2020 Phase 1 · MCQ (Single Correct Answer)

The product(s) of reaction catalyzed by nitrogenase in root nodules of leguminous plants is/are

(A) Ammonia alone
(B) Nitrate alone
(C) Ammonia and oxygen
(D) Ammonia and hydrogen

Open solution

Q13. NEET 2019 (Odisha) · MCQ (Single Correct Answer)

Which of the following bacteria reduce nitrate in soil into nitrogen?

(A) *Nitrosomonas* **(B)** *Nitrobacter* **(C)** *Nitrococcus* **(D)** *Thiobacillus*

Open solution

Q14. NEET 2018 · MCQ (Single Correct Answer)

Which of the following elements is responsible for maintaining turgor in cells?

(A) Sodium **(B)** Calcium **(C)** Magnesium **(D)** Potassium

Open solution

Q15. NEET 2018 · MCQ (Single Correct Answer)

In which of the following forms is iron absorbed by plants?

(A) Ferrous
(B) Both ferric and ferrous
(C) Ferric
(D) Free element

Open solution

Q16. NEET 2016 Phase 2 · MCQ (Single Correct Answer)

Which is essential for the growth of root tip?

(A) Zn **(B)** Fe **(C)** Ca **(D)** Mn

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q17. NEET 2016 Phase 1 · MCQ (Single Correct Answer)

In which of the following all three are macronutrients?

(A) Molybdenum, magnesium, manganese
(B) Nitrogen, phosphorus, sulphur
(C) Boron, Zinc, manganese
(D) Iron, copper, molybdenum

Open solution

Q18. AIPMT 2015 · MCQ (Single Correct Answer)

The oxygen evolved during photosynthesis, comes from water molecules. Which one of the following pairs of elements is involved in this reaction?

(A) Magnesium and Molybdenum
(B) Magnesium and Chlorine
(C) Manganese and chlorine
(D) Manganese and Potassium

Open solution

Q19. AIPMT 2015 · MCQ (Single Correct Answer)

During biological nitrogen fixation, inactivation of nitrogenase by oxygen poisoning is prevented by

(A) carotene
(B) cytochrome
(C) leghaemoglobin
(D) xanthophyll.

Open solution

Q20. AIPMT 2015 Cancelled Paper · MCQ (Single Correct Answer)

Minerals known to be required in large amounts for plant growth include

(A) potassium, phosphorus, selenium, boron
(B) magnesium, sulphur, iron, zinc
(C) phosphorus, potassium, sulphur, calcium
(D) calcium, magnesium, manganese, copper.

Open solution

Q21. AIPMT 2014 · MCQ (Single Correct Answer)

Deficiency symptoms of nitrogen and potassium are visible first in

(A) senescent leaves
(B) young leaves
(C) roots
(D) buds.

Open solution

Q22. NEET 2013 (Karnataka) · MCQ (Single Correct Answer)

Specialized cells for fixing atmospheric nitrogen in *Nostoc* are

(A) heterocysts
(B) hormogonia
(C) modules
(D) akinetes.

Open solution

Q23. NEET 2013 (Karnataka) · MCQ (Single Correct Answer)

Which of the following elements is a constituent of biotin?

(A) Magnesium
(B) Calcium
(C) Phosphorus
(D) Sulphur

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q24. NEET 2013 · MCQ (Single Correct Answer)

The first stable product of fixaation of atmospheric nitrogen in leguminous plants is

(A) NO_3^- (B) glutamate (C) NO_2^- (D) ammonia.

Open solution

Q25. AIPMT 2012 Mains · MCQ (Single Correct Answer)

For its action, nitrogenase requires

(A) high input of energy (B) light

(C) Mn^{2+} (D) super oxygen radicals.

Open solution

Q26. AIPMT 2012 Prelims · MCQ (Single Correct Answer)

Best defined function of manganese in green plants is

(A) photolysis of water (B) Calvin cycle

(C) nitrogen fixation (D) water absorption.

Open solution

Q27. AIPMT 2012 Prelims · MCQ (Single Correct Answer)

Which one of the following is correctly matched?

(A) Passive transport of nutrients – ATP (B) Apoplast – Plasmodesmata

(C) Potassium – Readily immobilisation (D) Bakane of rice seedlings – F. Skoog.

Open solution

Q28. AIPMT 2011 Mains · MCQ (Single Correct Answer)

Which one of the following is not an essential mineral element for plants while the remaining three are?

(A) Iron (B) Manganese (C) Cadmium (D) Phosphorus

Open solution

Q29. AIPMT 2011 Prelims · MCQ (Single Correct Answer)

Which one of the following elements in plants is not remobilised?

(A) Phosphorus (B) Calcium (C) Potassium (D) Sulphur

Open solution

Q30. AIPMT 2011 Prelims · MCQ (Single Correct Answer)

Nitrifying bacteria

(A) oxidize ammonia to nitrates

(B) convert free nitrogen to nitrogen compounds

(C) convert proteins into ammonia

(D) reduce nitrates to free nitrogen.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q31. AIPMT 2011 Prelims · MCQ (Single Correct Answer)

The function of leghaemoglobin in the root nodules of legumes is

(A) inhibition of nitrogenase activity
(B) oxygen removal
(C) nodule differentiation
(D) expression of *nif* gene.

Open solution

Q32. AIPMT 2011 Prelims · MCQ (Single Correct Answer)

Which one of the following helps in absorption of phosphorus from soil by plants?

(A) *Glomus*
(B) *Rhizobium*
(C) *Frankia*
(D) *Anabaena*

Open solution

Q33. AIPMT 2010 Mains · MCQ (Single Correct Answer)

Leguminous plants are able to fix atmospheric nitrogen through the process of symbiotic nitrogen through the process of symbiotic nitrogen fixation. which one of the following statements is not correct during this process of nitrogen fixation?

(A) Leghaemoglobin scavenges oxygen and is pinkish in colour.
(B) Nodules act as sites for nitrogen fixation.
(C) The enzyme nitrogenase catalyses the conversion of atmospheric N_2 to NH_3
(D) Nitrogenase is insensitive to oxygen.

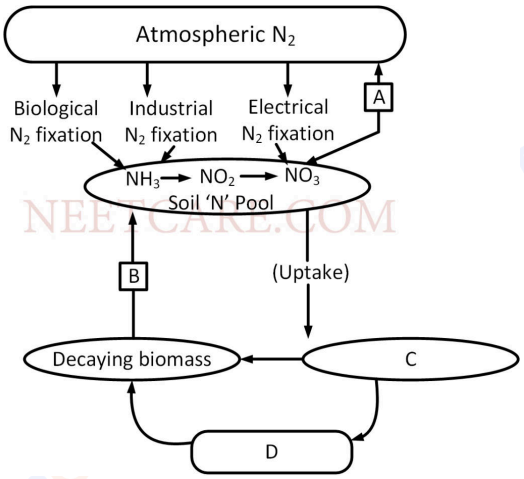
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q34. AIPMT 2010 Mains · MCQ (Single Correct Answer)

Study the cycle shown below and select the option which gives correct words for all the four blanks A, B, C and D.

Open solution



(A)

A	B	C	D
Nitrification	Ammonification	Animals	Plants

(B)

A	B	C	D
Denitrification	Ammonification	Plants	Animals

(C)

A	B	C	D
Nitrification	Denitrification	Animals	Plants

(D)

A	B	C	D
Denitrification	Nitrification	Plants	Animals

Q35. AIPMT 2010 Prelims · MCQ (Single Correct Answer)

An element playing important role in nitrogen fixation is

(A) molybdenum (B) copper (C) manganese (D) zinc.

Open solution

Q36. AIPMT 2010 Prelims · MCQ (Single Correct Answer)

Which one of the following is not a micronutrient?

(A) Molybdenum (B) Magnesium (C) Zinc (D) Boron

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q37. AIPMT 2009 · MCQ (Single Correct Answer)

Manganese is required in

(A) plant cell wall formation

(B) photolysis of water during photosynthesis

(C) chlorophyll synthesis

(D) nucleic acid synthesis.

Open solution

Q38. AIPMT 2007 · MCQ (Single Correct Answer)

Which one of the following elements is not an essential micronutrient for plant growth?

(A) Zn (B) Cu (C) Ca (D) Mn

Open solution

Q39. AIPMT 2007 · MCQ (Single Correct Answer)

A plant requires magnesium for

(A) protein synthesis (B) chlorophyll synthesis

(C) cell wall development (D) holding cells together.

Open solution

Q40. AIPMT 2006 · MCQ (Single Correct Answer)

Sulphur is an important nutrient for optimum growth and productivity in

(A) oilseed crops (B) pulse crops (C) cereals (D) fibre crops.

Open solution

Q41. AIPMT 2005 · MCQ (Single Correct Answer)

The deficiencies of micronutrients, not only affect growth of plants but also vital functions such as photosynthetic and mitochondrial electron flow. Among the list given below, which group of three elements shall affect most, both photosynthetic and mitochondrial electron transport?

(A) Co, Ni, Mo (B) Ca, K, Na (C) Mn, Co, Ca (D) Cu, Mn, Fe

Open solution

Q42. AIPMT 2004 · MCQ (Single Correct Answer)

If by radiation all nitrogenase enzyme are inactivated, then there will be no

(A) fixation of nitrogen in legumes

(B) fixation of atmospheric nitrogen

(C) conversion from nitrate to nitrite in legumes

(D) conversion from ammonium to nitrate in soil.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q43. AIPMT 2004 · MCQ (Single Correct Answer)

The most abundant element present in the plants is

(A) carbon (B) nitrogen (C) manganese (D) iron

Open solution

Q44. AIPMT 2003 · MCQ (Single Correct Answer)

Gray spots of oat are caused by deficiency of

(A) Cu (B) Zn (C) Mn (D) Fe

Open solution

Q45. AIPMT 2003 · MCQ (Single Correct Answer)

Boron in green plants assists in

(A) activation of enzymes (B) acting as enzyme co-factor
(C) photossynthesis (D) sugar transport.

Open solution

Q46. AIPMT 2003 · MCQ (Single Correct Answer)

The major portion of the dry weight of plants comprises of

(A) carbon, nitrogen and hydrogen (B) carbon, hydrogen and oxygen
(C) nitrogen, phosphorus and potassium (D) calcium, magnesium and sulphur

Open solution

Q47. AIPMT 2003 · MCQ (Single Correct Answer)

Which one of the following mineral elements plays an important role in biological nitrogen fixation?

(A) Zinc (B) Molybdenum (C) Copper (D) Manganese

Open solution

Q48. AIPMT 2002 · MCQ (Single Correct Answer)

Choose the correct match.
Bladderwort, sundew, Venus flytrap

(A) *Nepenthes, Dionea, Drosera* (B) *Nepenthes, Utricularia, Vanda*
(C) *Utricularia, Drosera, Dionea* (D) *Dionea, Trapa, Vanda.*

Open solution

Q49. AIPMT 2001 · MCQ (Single Correct Answer)

Roots of which plant contains a red pigment which have affinity for oxygen?

(A) Carrot (B) Soybean (C) Mustard (D) Radish

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q50. AIPMT 2001 · MCQ (Single Correct Answer)

Which aquatic fern performs nitrogen fixation?

(A) *Azolla* (B) *Nostoc* (C) *Salvia* (D) *Salvinia*

Open solution

Q51. AIPMT 2000 · MCQ (Single Correct Answer)

Plants take zinc in the form of

(A) $ZnSO_4$ (B) Zn^{++} (C) ZnO (D) Zn.

Open solution

Q52. AIPMT 2000 · MCQ (Single Correct Answer)

When the plants are grown in magnesium deficient but urea rich soil, the symptoms expressed are

(A) yellowish leaves (B) colourless petiole
(C) dark green leaves (D) shoot apex die.

Open solution

Q53. AIPMT 1998 · MCQ (Single Correct Answer)

Which of the following is non-symbiotic biofertilizer?

(A) VAM (B) *Azotobacter* (C) *Anabaena* (D) *Rhizobium*

Open solution

Q54. AIPMT 1997 · MCQ (Single Correct Answer)

Which of the following is not caused by deficiency of mineral nutrition?

(A) Etiolation (B) Shortening of internode
(C) Necrosis (D) Chlorosis

Open solution

Q55. AIPMT 1996 · MCQ (Single Correct Answer)

Which one of the following elements is almost non-essential for plants?

(A) Zn (B) Na (C) Ca (D) Mo

Open solution

Q56. AIPMT 1995 · MCQ (Single Correct Answer)

Which of the following elements plays an important role in biological nitrogen fixation?

(A) Copper (B) Molybdenum (C) Zinc (D) Manganese

Open solution

Q57. AIPMT 1994 · MCQ (Single Correct Answer)

Which of the following species does not have the ability to fix atmospheric nitrogen?

(A) *Azotobacter* (B) *Anabaena* (C) *Nostoc* (D) *Spirogyra*

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q58. AIPMT 1994 · MCQ (Single Correct Answer)

Which one of the following statements is correct?

(A) Legumes fix nitrogen only through the specialized bacteria that live in their roots.

(B) Legumes fix nitrogen independently of the specialized bacteria that live in their roots.

(C) Legumes fix nitrogen only through specialized bacteria that live in their leaves.

(D) Legumes are incapable of fixing nitrogen.

Open solution

Q59. AIPMT 1992 · MCQ (Single Correct Answer)

Amino acids are mostly synthesised from

(A) mineral salts

(B) fatty acids

(C) volatile acids

(D) α-ketoglutaric acid

Open solution

Q60. AIPMT 1991 · MCQ (Single Correct Answer)

Mineral associated with cytochrome is

(A) Cu

(B) Mg

(C) Cu and Mg

(D) Fe

Open solution

Q61. AIPMT 1989 · MCQ (Single Correct Answer)

Which one is an essential mineral, not constituent of any enzyme but stimulates the activity of many enzymes?

(A) Zn

(B) Mn

(C) K

(D) Mg

Open solution

Q62. AIPMT 1989 · MCQ (Single Correct Answer)

Phosphorous and nitrogen ions generally get depleted in soil because they usually occur as

(A) neutral ions

(B) negatively charged ions

(C) positively charged ions

(D) both positively and negatively charged but disproportionate mixture

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

1. **B**	2. **C**	3. **D**	4. **B**	5. **C**
6. **B**	7. **B**	8. **B**	9. **A**	10. **B**
11. **C**	12. **D**	13. **D**	14. **D**	15. **C**
16. **C**	17. **B**	18. **C**	19. **C**	20. **C**
21. **A**	22. **A**	23. **D**	24. **D**	25. **A**
26. **A**	27. **C**	28. **C**	29. **B**	30. **A**
31. **B**	32. **A**	33. **D**	34. **B**	35. **A**
36. **B**	37. **B**	38. **C**	39. **B**	40. **A**
41. **D**	42. **A**	43. **A**	44. **C**	45. **D**
46. **B**	47. **B**	48. **C**	49. **B**	50. **A**
51. **B**	52. **A**	53. **B**	54. **A**	55. **B**
56. **B**	57. **D**	58. **A**	59. **D**	60. **D**
61. **C**	62. **B**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs